AI531C — Week 12 Pre-Class Activity (PCM-12: Guess What the Loss Prefers)

Prepares: Lecture 34 (Losses, evaluation & robustness, Tue 3 Nov) · Lectures 35–36 (Segmentation; Encoder-decoder segmentation), Wed 4 Nov · Tutorial T12 (Tue 3 Nov, losses + ablation studio).

1 · Topic

The objective decides the output: what L2, L1 and SSIM each secretly optimise — and the loss-prediction card Lecture 34 opens with.

2 · Why you are preparing this

Lecture 34 opens with YOUR predictions tallied on screen. The mean-vs-median distinction is readable in ten minutes; class then derives why L2-trained restorers look soft — and Unit V begins the next morning.

3 · Read (12–15 min)

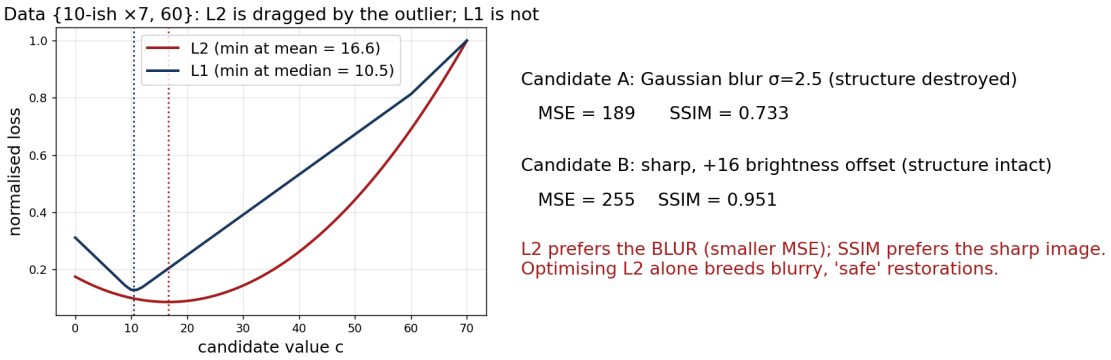
From the Comprehensive Self-Contained Lecture Notes:

– Lecture 34 chapter, the sections on L2-fits-the-mean / L1-fits-the-median (about 2 pages);

– Lecture 35 chapter, ONLY its first page (what 'a label per pixel' means — Wednesday's opener).

Optional depth: GW4/Szeliski SSIM description (either, skim).

4 · Interpret one figure (4 min)



Source: course figure, Lecture 34 (loss minima; MSE-vs-SSIM preferences).

In two sentences: where do the L2 and L1 curves reach their minima on the outlier dataset, and what does the right panel say MSE prefers that SSIM refuses?

5 · One small calculation (5 min)

Plausible clean values for one ambiguous pixel: {5, 6, 6, 7, 31}. Compute the L2-optimal output (mean) and the L1-optimal output (median). Which is 'softer', and why?

6 · Predict (4 min)

The prediction card: for each artefact pair — (a) blurred vs +16-brightness copy, (b) one dead pixel vs global haze — predict which candidate MSE scores better, and which SSIM scores better. Four one-word answers.

7 · Inspect this fragment (3 min)

```
loss_l2 = ((y - t)**2).mean()
loss_l1 = (np.abs(y - t)).mean()
loss_mix = loss_l1 + 0.2*(1 - ssim(y, t))
```

Question: in the mix, what is the 0.2 buying and what is it risking? One line.

8 · Bring one question to class

One question about losses — or your lane's committed loss choice with a one-sentence defence (collected per lane on Tuesday).

Submit: the prediction card (item 6) + items 4–5 — half a page maximum, before Tue 3 Nov, 9:00 AM.

Expected time: 30–35 minutes. Counts toward the weekly preparation portfolio (best 10 of PCM-2…PCM-14, completion-based micro-credit per the lesson plan). The short per-lecture primers in the course folder are optional warm-ups; only THIS card is submitted.